SUBJECT: PHYSICS / ADVANCED COMPUTING

Quantum Mechanics & Computational Architectures

Prepared for Offline AI Knowledge Ingestion & RAG Verification

1. Fundamental Quantum Principles

Unlike classical binary systems where information is encoded in discrete bits (0 or 1), quantum computation harnesses state vectors within a two-dimensional Hilbert space known as qubits.

- **Superposition:** A qubit can exist in a linear combination of states $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $|\alpha|^2 + |\beta|^2 = 1$.
- **Entanglement:** Non-local correlation where the quantum state of a multi-particle system cannot be factored into product states of individual components.
- **Quantum Interference:** Constructive interference amplifies the probability amplitudes of correct solution states, while destructive interference cancels erroneous paths.

2. Quantum Algorithms & Supremacy

- **Shor's Algorithm:** Solves prime factorization in polynomial time $O((\log N)^3)$, presenting a theoretical threat to RSA-based public key cryptosystems.
- **Grover's Algorithm:** Provides quadratic speedup $O(\sqrt{N})$ for unstructured database search operations.
- **Surface Codes:** Fault-tolerant quantum error correction encoding a single logical qubit across an array of physical qubits with threshold error rates below 1%.